

ERT 474/574 LAB #1

Course Platform Setup



Outline

- Set up your **GitHub repo** in the **GitHub organization**
 - Step-by-step instructions
- Get familiar with the **GitHub CodeSpace**
 - The coding interface – Microsoft VS Code
 - Jupyter Notebook
 - Terminal (folder structures)
- Course logistics
 - How to do future lab sessions?
 - How to submit homework?

By the end of this lab, you will need to know

1. Open ***Github CodeSpace***
2. Pull updates from the ***CourseMaterials26*** repo
 - Materials in labs will be published in ***CourseMaterials26***
3. Navigate yourself in VSCode, including creating folders and notebooks
4. Submit homework by ***pushing changes*** to your ***homework repo***

Organization

- <https://github.com/OS-Hydro-Analytics-Fall-2026>



Go to https://github.com/OS-Hydro-Analytics-Fall-2026/homework_template

The screenshot shows the GitHub repository page for 'homework_template' under the organization 'OS-Hydro-Analytics-Fall-2026'. The repository is a public template. A callout box highlights the 'Use this template' button, which has a dropdown menu with the following options: 'Create a new repository' and 'Open in a codespace'.

Repository Details:

- Repository: homework_template (Public template)
- Watch: 0, Fork: 0, Star: 0
- Buttons: Watch, Fork, Star, Use this template (dropdown), Create a new repository, Open in a codespace

Files and Commits:

File	Commit Message	Commit Hash	Time Ago
act-hydro	update devcontainer and postbuild.sh	4f04b72	3 days ago
.devcontainer	update devcontainer and postbuild.sh		3 days ago
homework_submissions	Add devcontainer stuff and scaffold		last year
project_submissions	add project_submission		4 days ago
.gitignore	Initial commit		last year
README.md	Initial commit		last year
postbuild.sh	update devcontainer and postbuild.sh		3 days ago
pyproject.toml	update python packages		4 days ago

About: No description, website, or topics provided.

Readme: (Link to README file)

Releases: No releases published

Packages: No packages published

Click “Create a new repository”

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

Start with a template

Templates pre-configure your repository with files.

 OS-Hydro-Analytics-Fall- 2026/homework_template ▾

Include all branches

If enabled, all branches from the template repository will be included.

Off ☐

1 General

Owner *

 OS-Hydro-Analytics-Fall-2026 ▾

Repository name *

homework_YourUserName

✓ homework_YourUserName is available.

Great repository names are short and memorable. How about **fantastic-sniffle?**

Description

0 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

 Private ▾

Create repository

Make sure to select
“OS-Hydro-Analytics-
Fall-2026”

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

Start with a template

Templates pre-configure your repository with files.

 OS-Hydro-Analytics-Fall- 2026/homework_template ▾

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Great repository names are short and memorable. How about **fantastic-sniffle?**

Description

0 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

 Private ▾

Create repository

Make sure to select
“OS-Hydro-Analytics-
Fall-2026”

Name your repo as
“homework_YourUser
Name”

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

Start with a template

Templates pre-configure your repository with files.

 OS-Hydro-Analytics-Fall- 2026/homework_template ▾

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 homework_YourUserName is available.

Great repository names are short and memorable. How about **fantastic-sniffle?**

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0 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

 Private ▾

Create repository

Make sure to select
“OS-Hydro-Analytics-
Fall-2026”

Name your repo as
“homework_YourUser
Name”

Choose visibility
“Private”

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

Start with a template

Templates pre-configure your repository with files.

 OS-Hydro-Analytics-Fall- 2026/homework_template ▾

Include all branches

If enabled, all branches from the template repository will be included.

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1 General

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 OS-Hydro-Analytics-Fall-2026 ▾

Repository name *

homework_YourUserName

 homework_YourUserName is available.

Great repository names are short and memorable. How about **fantastic-sniffle?**

Description

0 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

 Private ▾

Create repository

Make sure to select
“OS-Hydro-Analytics-
Fall-2026”

Name your repo as
“homework_YourUser
Name”

Choose visibility
“Private”

Smash “Create repo”

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

Start with a template

Templates pre-configure your repository with files.

 OS-Hydro-Analytics-Fall- 2026/homework_template ▾

Include all branches

If enabled, all branches from the template repository will be included.

Off ☐

1 General

Owner *

 OS-Hydro-Analytics-Fall-2026 ▾

Repository name *

homework_YourUserName

 homework_YourUserName is available.

Great repository names are short and memorable. How about **fantastic-sniffle**?

Description

0 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

 Private ▾

Create repository

By now, you should have your own homework directory

https://github.com/OS-Hydro-Analytics-Fall-2026/homework_YourUserName

Do NOT capitalize “h”

You will need to replace the highlighted part using your own username

GitHub CodeSpaces



The screenshot shows a GitHub repository named 'homework_yifan' (Private) under the organization 'OS-Hydro-Analytics-Fall-2026'. A notification at the top states 'Codespace "glowing garbanzo" deleted.' The repository is generated from 'OS-Hydro-Analytics-Fall-2026/homework_template'. The file browser shows a 'main' branch with 1 branch and 0 tags. The file list includes:

- .devcontainer (Initial commit)
- homework_submissions (test of commit hom)
- project_submissions (Initial commit)
- .gitignore (Initial commit)
- README.md (Initial commit)
- postbuild.sh (Initial commit)
- pyproject.toml (Initial commit)

 A 'README' link is at the bottom. A 'Code' button is highlighted in green. An overlay titled 'Codespaces' is centered, showing 'No codespaces' and a green button labeled 'Create codespace on main'. The 'About' section on the right notes 'No description, website, or topics provided.' and lists 'Readme' and 'Activity'. The 'Releases' section states 'No releases published' with a link to 'Create a new release'. The 'Packages' section is empty. At the bottom of the overlay, it says 'Codespace usage for this repository is paid for by YifanCheng.'

Smash “Create codespace on main” to create a CodeSpace

The screenshot displays the VS Code Web interface. On the left, the **Side Bar** (labeled in a yellow box) contains the **EXPLORER** view, showing a workspace named **WORKSPACES [CODESPACES: SUPER FIESTA]** with files like `.codespaces`, `.venv`, `CourseMaterials25`, `homework_yifan`, `pyproject.toml`, and `uv.lock`. The main **Editor** area (labeled in a yellow box) features a 'Get Started with VS Code for the Web' walkthrough. It includes a 'Choose your theme' section with a 'Browse Color Themes' button and a tip about keyboard shortcuts. Four theme preview cards are shown: **Dark Modern**, **Light Modern**, **Dark High Contrast**, and **Light High Contrast**. A 'See More Themes...' link is also present. The bottom **Panel** (labeled in a yellow box) shows the **TERMINAL** view with the following output:

```
+ websocket-client==1.8.0
+ widgetsnbextension==4.0.14
+ xarray==2025.8.0
+ xyzservices==2025.4.0
+ yarl==1.20.1
+ zarr==3.1.2
+ zipp==3.23.0
Cloning into 'CourseMaterials25'...
remote: Enumerating objects: 36, done.
remote: Counting objects: 100% (36/36), done.
remote: Compressing objects: 100% (27/27), done.
remote: Total 36 (delta 5), reused 31 (delta 3), pack-reused 0 (from 0)
Receiving objects: 100% (36/36), 5.41 MiB | 27.43 MiB/s, done.
Resolving deltas: 100% (5/5), done.
Outcome: success User: vscode WorkspaceFolder: /workspaces
devcontainer process exited with exit code 0
Finished configuring codespace.
```

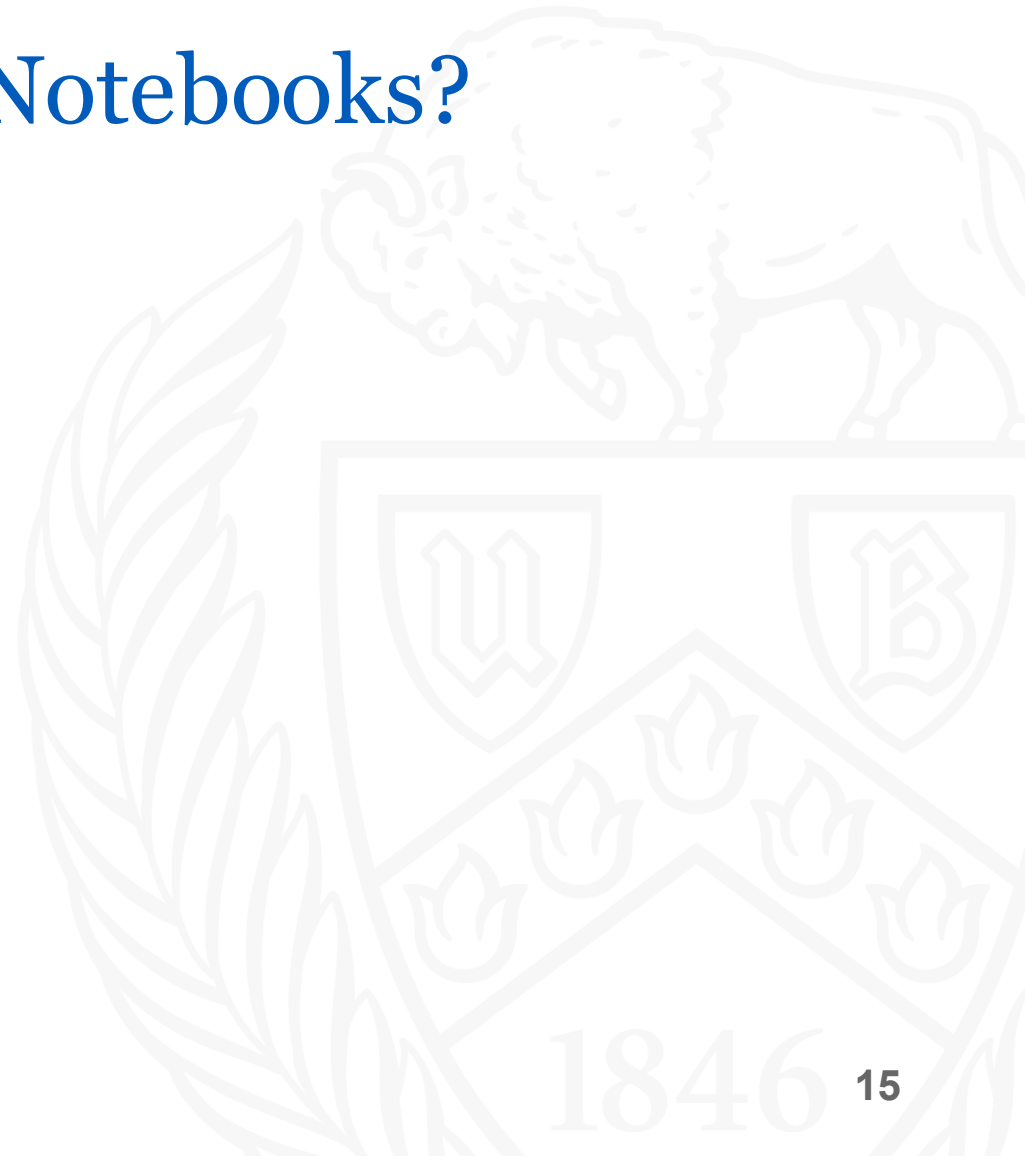
The bottom status bar shows the current workspace as **Codespaces: super fiesta**, the file `homework_yifan`, and the branch `main`. It also includes icons for search, run and debug, and a user profile icon.

The screenshot shows the VS Code Web interface with several annotations in yellow boxes:

- Side Bar:** Points to the left sidebar containing the Explorer, Search, Source Control, Run and Debug, Extensions, Testing, and Accounts views.
- Editor:** Points to the main editor area displaying the "Get Started with VS Code for the Web" walkthrough.
- Panel:** Points to the bottom panel showing the "TERMINAL" view with output logs.
- Click + to open new terminals:** Points to the "+" button in the terminal panel header.

The interface includes a top bar with a search bar and workspace name, a left sidebar with navigation icons, a main editor area with a walkthrough, and a bottom panel with a terminal view.

How to create folders & Jupyter Notebooks?



The screenshot displays the Visual Studio Code editor interface. On the left, the Explorer sidebar shows a file tree with folders like `.codespaces`, `.venv`, `CourseMaterials`, `homework1`, `project_submissions`, and `uv.lock`. A yellow callout box with the text "Click this to create new folder" points to the "New Folder" icon (a folder with a plus sign) in the Explorer sidebar. The main editor area shows a Jupyter Notebook with three cells. The first cell contains `%matplotlib inline`. The second cell contains import statements: `import numpy as np`, `import pandas as pd`, `import xarray as xr`, and `import matplotlib.pyplot as plt`. The third cell contains `print(1+2)`. The output of the third cell is `3`. At the bottom, the TERMINAL panel shows a bash prompt: `@YifanCheng → /workspaces $`. The status bar at the bottom indicates "Spaces: 4", "Finish Setup", "Cell 1 of 4", and "Layout: U.S."

workspaces [Codespaces: super fiesta]

EXPLORER

WORKSPACES [CODESPACES...]

.codespaces

.venv

CourseMaterials

homework1

project_submissions

uv.lock

Click this to create new folder

homework1.ipynb

Generate + Code + Markdown | Run All | Clear All Outputs | Outline

Select Kernel

Python

```
%matplotlib inline
```

```
import numpy as np
import pandas as pd
import xarray as xr
import matplotlib.pyplot as plt
```

[2]

```
print(1+2)
```

[3]

3

Python

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS 2

bash

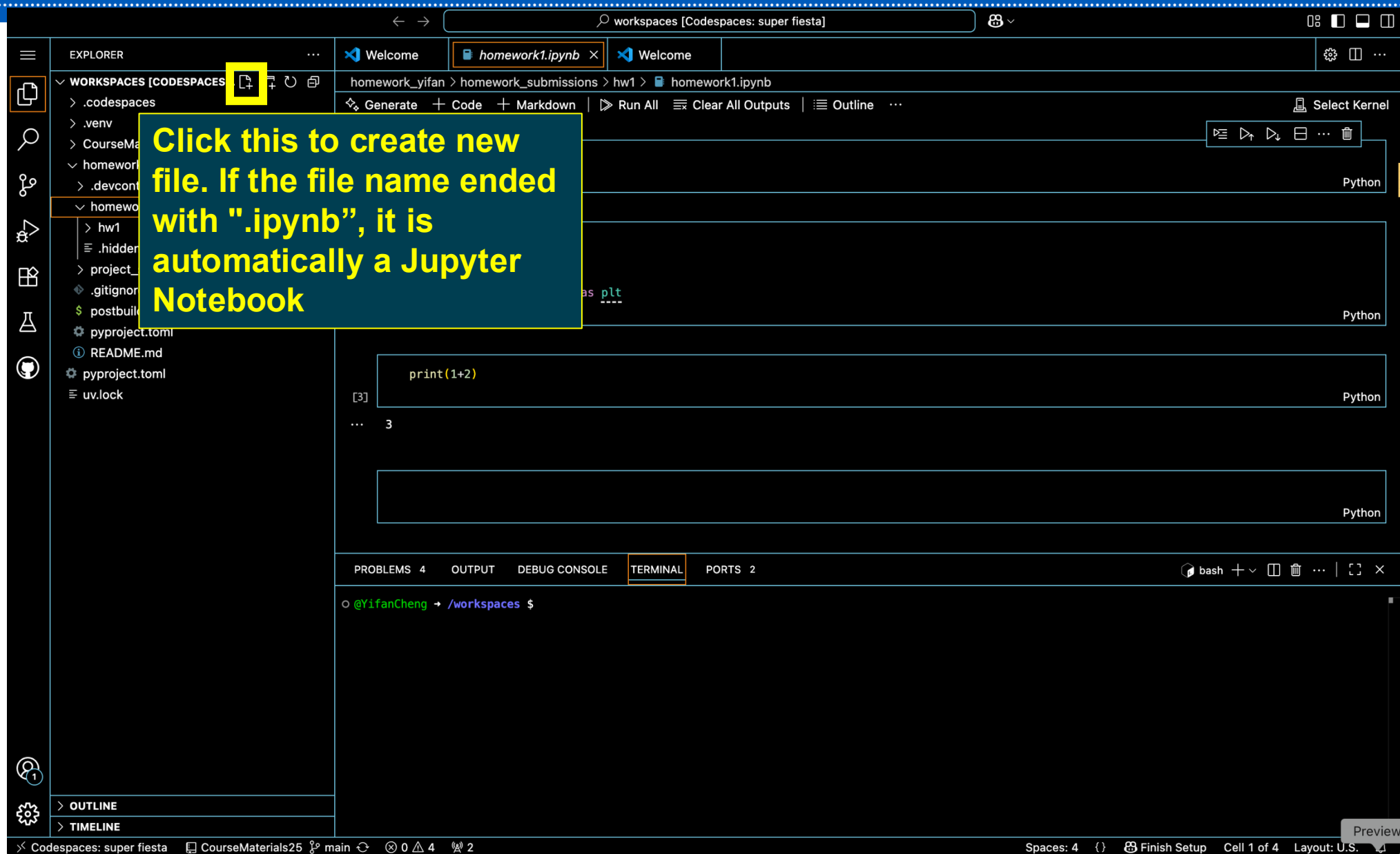
@YifanCheng → /workspaces \$

OUTLINE

TIMELINE

Codespaces: super fiesta CourseMaterials25 main 0 4 2

Spaces: 4 Finish Setup Cell 1 of 4 Layout: U.S.



Click this to create new file. If the file name ended with ".ipynb", it is automatically a Jupyter Notebook

```
print(1+2)
```

[3]
... 3

PROBLEMS 4 OUTPUT DEBUG CONSOLE **TERMINAL** PORTS 2

@YifanCheng → /workspaces \$

Spaces: 4 {} Finish Setup Cell 1 of 4 Layout: U.S.

How to open a Notebook and how to run a Jupyter Notebook?

1. Click filename to open the notebook

2. Click "Select Kernel"

3. Click "Python Environments"

```
import numpy as np
import pandas as pd
import xarray as xr
import matplotlib.pyplot as plt
```

```
def add_two_numbers(a,b):
    return a+b
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE **TERMINAL** PORTS 3

@YifanCheng -> /workspaces \$

Select a Python Environment

- Create Python Environment
- ★ ert574 (Python 3.11.13) .venv/bin/python **Recommended**
- Python 3.11.13 /usr/local/bin/python **Global Env**
- Python 3.11.2 /bin/python3
- Python 3.11.2 /usr/bin/python3

Click "ert574"

This is the Python Virtual Environment that I specifically designed for this class.

```
def add_two_numbers(a,b):  
    return a+b
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE **TERMINAL** PORTS 3

@YifanCheng → /workspaces \$

Spaces: 4 {} Finish Setup Cell 2 of 5 Layout: U.S.

The screenshot displays the JupyterLab environment within a web browser. The top navigation bar shows the workspace name "workspaces [Codespaces: super fiesta]". The left sidebar contains the Explorer view, showing a file tree for the workspace. The main area displays a Jupyter notebook with several code cells. The first cell, containing the code `%matplotlib inline`, is highlighted with a yellow border. A text overlay with a yellow background and black text reads: "This is a code cell" and "You will be able to run each cell individually". Below this, the second cell contains a Python function definition: `def add_two_numbers(a,b):` followed by an indented `return a+b`. The bottom of the interface shows the Terminal view with a bash prompt and the current directory `/workspaces`.

workspaces [Codespaces: super fiesta]

EXPLORER

WORKSPACES [CODESPACES: SUPER FIESTA]

- > .codespaces
- > .venv
- > CourseMaterials25
- ▼ homework_yifan
 - > .devcontainer
 - ▼ homework_submissions
 - ▼ hw1
 - homework1.ipynb
- ▼ project_submissions
- ▼ .gitignore
- \$ postbuild.sh
- ⚙ pyproject.toml
- 📖 README.md
- ⚙ pyproject.toml
- ≡ uv.lock

homework_yifan > homework_submissions > hw1 > homework1.ipynb > empty cell

Generate + Code + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline ...

ert574 (Python 3.11.13)

[1] ✓ 8.5s

This is a code cell

You will be able to run each cell individually

[2] Python

[3] ✓ 0.0s

... 3

[4] ✓ 0.2s

Python

def add_two_numbers(a,b):
| return a+b

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 9 JUPYTER

@YifanCheng → /workspaces \$

Spaces: 4 Finish Setup Cell 5 of 5 Layout: U.S.

When moving your cursor to this cell, you can notice there is a triangle button popping up. Click that to run this cell.

Practice #1

- Create a folder named “practice” under the folder “homework_submissions”
- Within the “practice” folder, create a Jupyter Notebook named “practice.ipynb”
- In “practice.ipynb”,
 - In one cell, write a function that adds up two variables, **a** and **b**, and assigns the sum to the variable **result**. You may seek help from *CoPilot*.
 - In another cell, assign **a**=3, **b**=2, print the sum of **a** and **b**



How can we “Submit” the homework?

“Submitting” means that your instructor will be able to access the Jupyter Notebook that you worked on.

We will need to

1. “commit” the changes locally in GitHub CodeSpaces
2. “push” the committed changes to GitHub

workspaces [Codespaces: super fiesta]

EXPLORER

WORKSPACES [CODESPACES: SUPER FIESTA]

- > .codespaces
- > .venv
- > CourseMaterials25
- homework_yifan
- .devcontainer
- homework_submissions
- hw1
 - homework1.ipynb M
 - .hidden
 - project_submissions
 - .gitignore
 - postbuild.sh
 - pyproject.toml
 - README.md
 - pyproject.toml
 - uv.lock

homework_yifan > homework_submissions > hw1 > homework1.ipynb > empty cell

Generate + Code + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline ...

ert574 (Python 3.11.13)

[1] ✓ 8.5s

```
%matplotlib inline
```

Python

[2] ✓ 5.3s

```
import numpy as np
import pandas as pd
import xarray as xr
import matplotlib.pyplot as plt
```

Python

Python

[4] ✓ 0.2s

```
return a+b
```

Python

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 9 JUPYTER

@YifanCheng → /workspaces \$

Spaces: 4 Finish Setup Cell 5 of 5 Layout: U.S.

The screenshot shows a JupyterLab workspace with a dark theme. On the left, the 'SOURCE CONTROL' sidebar is open, showing a list of changes. Two changes are visible: 'homework1.ipynb' and 'homework1.ipynb'. The 'homework1.ipynb' change is highlighted with a yellow circle. The main editor area shows a Jupyter notebook with two cells. The first cell contains the code `print(1+2)` and has been executed, showing the output `3`. The second cell contains the code `def add_two_numbers(a,b):` and has been executed, showing the output `0.2s`. A yellow text box is overlaid on the screenshot, containing the text: 'We will need to commit the changes we made to the files! Type "add practice" in the Message box, and click "Commit"'. The bottom status bar shows 'Spaces: 4', 'Finish Setup', 'Cell 6 of 6', and 'Layout: U.S.'

We will need to commit the changes we made to the files!

Type "add practice" in the Message box, and click "Commit"

workspaces [Codespaces: super fiesta]

SOURCE CONTROL

CHANGES

CourseMat main

Message (%Enter to commit on "...)

Commit

homework main

Message (%Enter to commit on "...)

Commit

Changes

homework1.ipynb

test of commit homework1.ipynb

Initial commit YifanChen

homework1.ipynb M

Generate + Code + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline

ert574 (Python 3.11.13)

Python

Python

Python

Python

bash

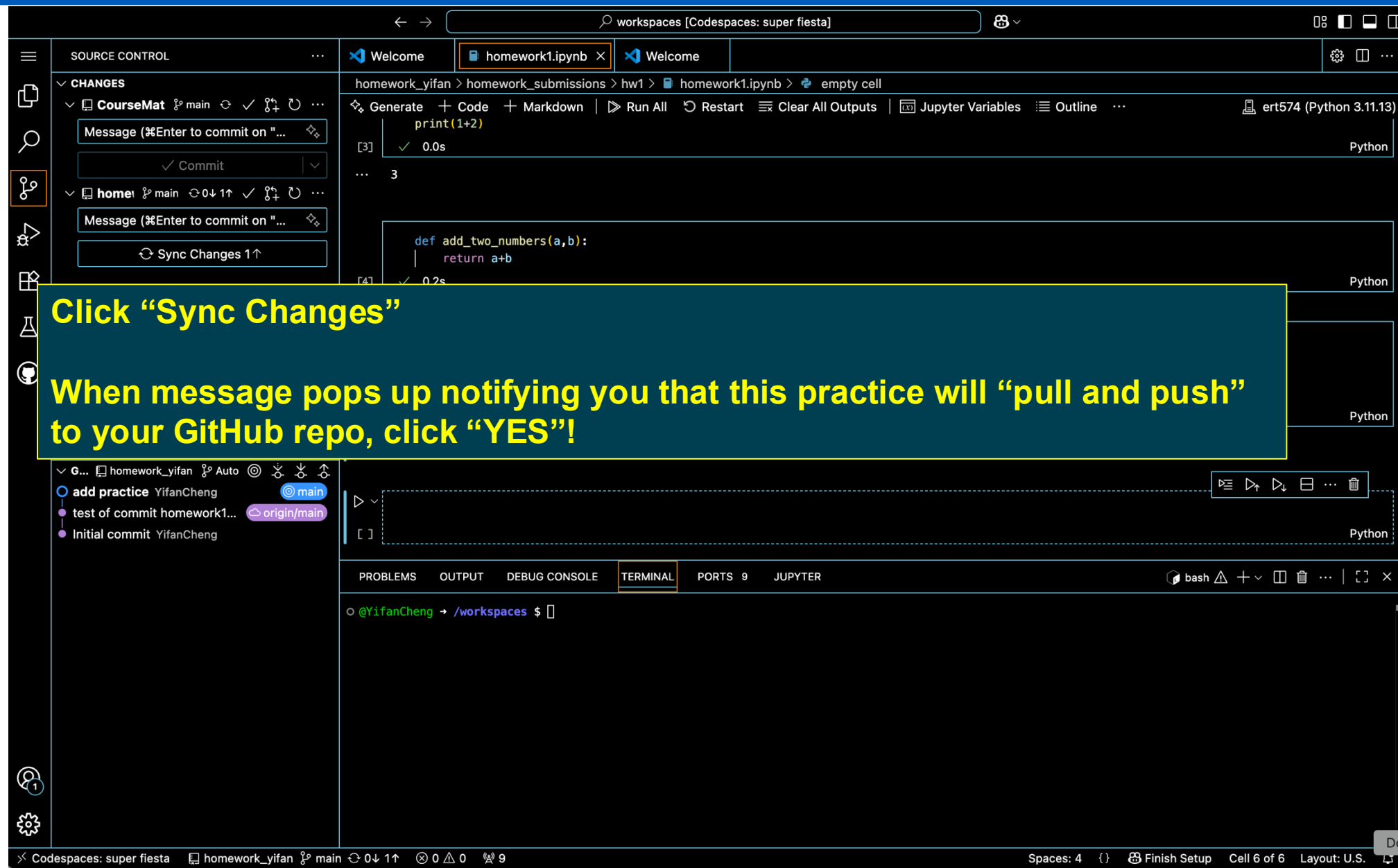
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 9 JUPYTER

Codespaces: super fiesta homework_yifan main 0 0 9 Spaces: 4 {} Finish Setup Cell 6 of 6 Layout: U.S.

27

After you clicked "Commit", a warning message pops up! This means that we combined two steps "stage change" and "commit change" into one step. Please click "Yes" there.

After we commit the changes, it means that the changes has been documented locally in CodeSpaces. We still need to “push” the changes to GitHub repo.

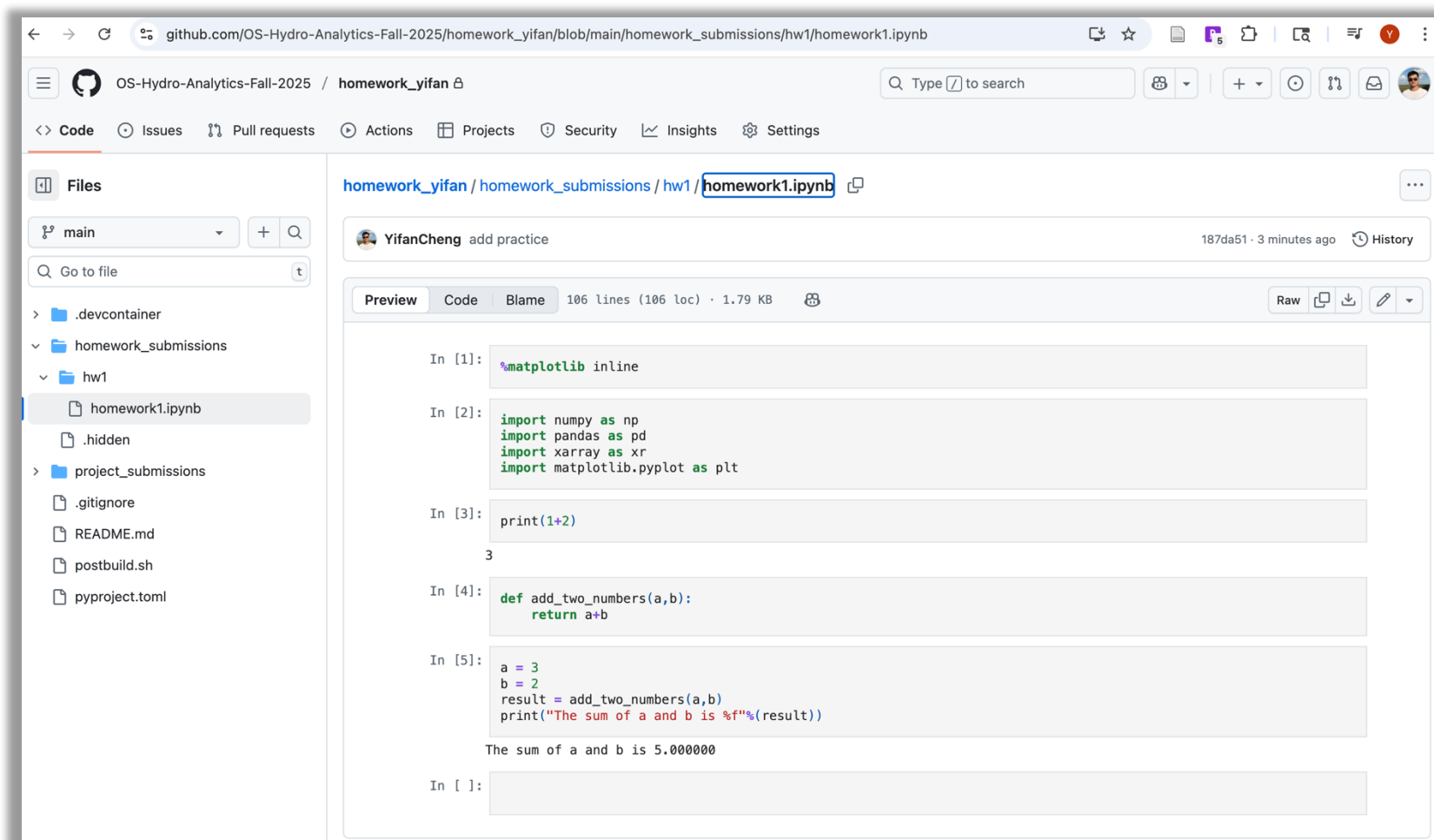


The screenshot shows a JupyterLab workspace titled "workspaces [Codespaces: super fiesta]". The interface includes a left sidebar with "SOURCE CONTROL" and "CHANGES" sections. The "CHANGES" section shows a commit message "Message (¶Enter to commit on \"...\"") and a "Commit" button. The main editor area displays a Jupyter notebook with two code cells. The first cell contains `print(1+2)` and the second cell contains `def add_two_numbers(a,b):` followed by `return a+b`. The bottom status bar shows the terminal with the prompt `@YifanCheng → /workspaces $`.

Click “Sync Changes”

When message pops up notifying you that this practice will “pull and push” to your GitHub repo, click “YES”!

Now, go to your homework repo



github.com/OS-Hydro-Analytics-Fall-2025/homework_yifan/blob/main/homework_submissions/hw1/homework1.ipynb

OS-Hydro-Analytics-Fall-2025 / homework_yifan

Code Issues Pull requests Actions Projects Security Insights Settings

Files

- main
- Go to file
- .devcontainer
- homework_submissions
 - hw1
 - homework1.ipynb
- project_submissions
 - .gitignore
 - README.md
 - postbuild.sh
 - pyproject.toml

homework_yifan / homework_submissions / hw1 / homework1.ipynb

YifanCheng add practice 187da51 · 3 minutes ago History

Preview Code Blame 106 lines (106 loc) · 1.79 KB Raw

```
In [1]: %matplotlib inline

In [2]: import numpy as np
import pandas as pd
import xarray as xr
import matplotlib.pyplot as plt

In [3]: print(1+2)
3

In [4]: def add_two_numbers(a,b):
return a+b

In [5]: a = 3
b = 2
result = add_two_numbers(a,b)
print("The sum of a and b is %f"%(result))

The sum of a and b is 5.000000

In [ ]:
```

You should see that the edits you made in the Jupyter Notebooks have been updated to GitHub Page!

Practice #2

- Go to your CodeSpaces, find the “**README**” file under your own homework folder (the folder should have the name, i.e., “homework_YourUserName”)
- Edit the **README** file to include the following information
 - 1. Name
 - 2. Email address
 - 3. Your coding experiences
- After making the changes, push the changes to your GitHub repo.

